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Color kinetics of longan flesh drying at high temperature

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Abstract

The aim of this study was to investigate the color kinetics during hot air drying of longan flesh at high temperatures. The longan of E-dor variety (*Dimocarpus Longan Lour.*) was used in this study. The sample sizes of 25–30 mm in diameter with the initial moisture content of 470–510% dry basis were tested. The experiments were carried out in the laboratory dryer at the drying air temperature of 60–130°C. Kinetic parameters for the color change were determined using CIE system measuring L^* , a^* and b^* values. These values were used to calculate the total color difference (ΔE), chroma and hue angle. The data of L^* , a^* , b^* and ΔE were fitted to various kinetic models. It was found that the modified kinetic model (second order polynomial equation) provided the best fit of the data of L^* and b^* . The zero-order kinetic model was found to be the most suitable for describing the a^* data and this model was found to be effective for representing the changes in ΔE values. Both models could predict the color change during drying relatively well. The simulated results showed that the drying air temperature and drying time affected the color change during the drying processes. The value of L^* decreased when the drying time and temperature increased. The values of a^* , b^* and ΔE increased in the first period of drying afterwards these values decreased with drying time and drying air temperature.

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1. Introduction

Longan is one the most economically important fruits in the northern part of Thailand. There are several preservation methods for reducing spoilage, increasing shelf life and adding value to product, such as canning, freezing and drying. The export value of dried longan in 2010 was \$68 million [1]. The quality of dried product is a specific characteristic that is acceptable by the consumer. The color is one of the most important quality factors of dried longan flesh. Browning reaction occurs during the drying of the fruit and this has a significant impact on the color of the final dried product. The rate of the browning

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reaction increases with increasing product temperature and the increase is faster when the product contains high sugar content [2]. Browning reactions can occur during drying of longan flesh because longan flesh is rich in sugar [3]. The kinetic models of color change are essential to design a new process for achieving the best quality of dried product. Several researchers studied on the kinetics of color changes of vegetables and fruits such as banana, apple, carrot and potato [4], kiwifruit [5], spinach [6] pineapple puree [7], whole longan [8] and longan without stone [9] but no kinetics studied related with the color change during drying of longan flesh were found. Therefore, the aim of this research is to study the kinetics of color change of longan flesh during drying in order to predict color changes with time and drying air temperature.

Nomenclature

L^*, a^*, b^*	color parameters
C	concentration of a quality factor
C_0, C_t	values of color parameters at time zero and t
k_0, k_1, A, B	kinetic constants of color parameters (min^{-1})
a_1, a_2, a_3, a_4, a_5	constant parameters
P	kinetic parameter
ΔE	total color difference
n	order of reaction
t	time (min)
T	temperature ($^{\circ}\text{C}$)

2. Materials and methods

2.1 Experiment set-up

A schematic diagram of a hot air dryer is shown in Fig. 1. The dryer mainly consisted of a 15 kW electric heater, a drying chamber with dimensions of $450 \times 450 \times 450 \text{ mm}^3$ and a backward curved blade centrifugal fan driven by a 0.75 kW motor.

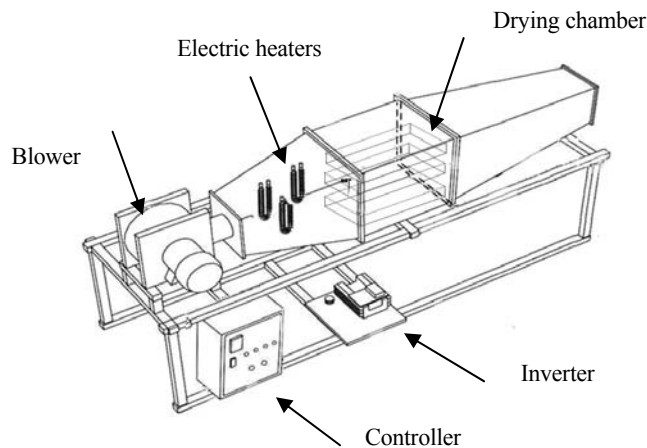


Fig. 1. Schematic diagram of hot air dryer

2.2 Material

Whole longan fruits of E-dor variety (*Dimocarpus Longan Lour.*) were purchased from a local market in Chiang Mai province. The samples, grade AA (25-30 mm in diameter) are peeled and punched at the pole to remove the stone before testing. The initial moisture content of sample was determined by hot air oven using temperature of 103°C, drying for 72 h and found to be 470 - 510% dry basis.

2.3 Drying experiment

The experiments were conducted at drying temperatures of 60, 80, 100, 110, 120 and 130°C with a constant air velocity of 0.7 m/s. In each drying experiment, 150 longan fruits were placed on three trays (50 fruits per tray), and the trays were stacked in the dryer. During drying, the five samples were removed from the dryer to measure the color at one hour intervals during the first three hours and at two hour intervals afterward until the final moisture content of the samples was 20% dry basis. In each period of time, the color of longan was measured at three different positions. Three measurements were taken at each position and the average of forty five measurements of each color parameter (L^* , a^* , b^*) was reported.

2.4 Color measurement

Color was measured with a colorimeter (Hunter Associates Laboratory, Inc., model MiniScan XE Plus, VA, USA). The equipment was calibrated before the measurement with white tile ($X = 79.9$, $Y = 84.4$, $Z = 90.4$). The color system used was the CIE (Commission International de l'Éclairage) where L^* represents lightness or darkness, a^* represents redness or greenness and b^* represents yellowness or blueness. In addition, the total color difference (ΔE), chroma, which indicates the purity or saturation of the color and hue angle which expresses the color change (an angle of 0 or 360° represents red hue, while angles of 90, 180, and 270° indicate yellow, green and blue hue, respectively) were calculated as follows[10]:

$$\Delta E = \sqrt{(L_0^* - L_t^*)^2 + (a_0^* - a_t^*)^2 + (b_0^* - b_t^*)^2} \quad (1)$$

$$\text{Chroma} = \sqrt{a_t^{*2} + b_t^{*2}} \quad (2)$$

$$\text{Hue angle} = \tan^{-1} \left(\frac{b_t^*}{a_t^*} \right) \quad (3)$$

2.5 Calculation of kinetic parameters

In order to understand how the color change of food materials as a function of drying time, the general equation that describes the reaction rate is given by [11],[12];

$$\frac{dC}{dt} = -k(C)^n \quad (4)$$

Where C is the concentration of a quality factor C at time t , k is the reaction rate constant and n is the order of reaction. By integration Eq.(4), the zero-order and first-order kinetic models are represented in Eq. (5) and (6), respectively. Also a modified kinetic model is applied to study the color change of food as in Eq.(7)

$$\text{Model 1: } C_t - C_0 = -k_0 \cdot t \quad (5)$$

$$\text{Model 2: } \frac{C_t}{C_0} = \exp(-k_1 \cdot t) \quad (6)$$

$$\text{Model 3: } \frac{C_t}{C_0} = A \cdot t^2 + B \cdot t + I \quad (7)$$

Where C_0 represents the initial value of color parameters (L^* , a^* , b^*) at time zero, C_t is the value at time t , and k_0 , k_1 , A , B are the kinetic parameters. The empirical equation which is used to describe the relationship between the kinetic parameters and the temperature is

$$P = a_1 T^4 + a_2 T^3 + a_3 T^2 + a_4 T + a_5 \quad (8)$$

Where P is kinetic parameter (k_0 , k_1 , A , B) and a_1 , a_2 , a_3 , a_4 , a_5 are the constant parameters.

The parameters (k_0 , k_1 , A , B , a_1 , a_2 , a_3 , a_4 , a_5) are determined by fitting the experimental data to Eqs. (5)-(8) using least square method.

3. Results and discussion

3.1 Color parameters

The color changes in longan flesh drying at different drying air temperatures are shown in Fig. 2. As shown in Fig. 2(a), at low drying air temperatures (60-100°C), L^* values increased slowly in the first 180 min of drying and then decreased afterwards. These values decreased as drying air temperature increased. At high drying air temperatures (110-130°C), the L^* values decreased quickly after the first 60 min of drying and then decreased. These values also decreased when drying air temperature increased. Fig. 2(b) shows that the a^* values increased during drying for all drying air temperatures and increased as drying air temperature increased. As seen in Fig. 2(c), at low temperatures, b^* values increased and then decreased with increasing drying time. The b^* values increased after 60 min of drying afterwards decreased quickly with increasing drying time at high temperatures. The effect of drying air temperature on the changes of b^* values was not obvious. These results indicate that the browning reaction could occur during drying. The total color difference (ΔE), chroma and hue angle were calculated using Eqs.(1)-(3). Fig. 2(d)-Fig. 2(f) show the total color difference (ΔE), chroma and hue angle, respectively. The total color difference increased while hue angle decreased with increasing drying time and drying air temperature. The chroma increased and then decreased with increasing drying time at low temperatures. At high drying air temperatures, the chroma increased after 60 min of drying afterwards decreased quickly with increasing drying time. This study show that characteristics of color changes of longan flesh during drying, which showed tendency change from yellow to red hues.

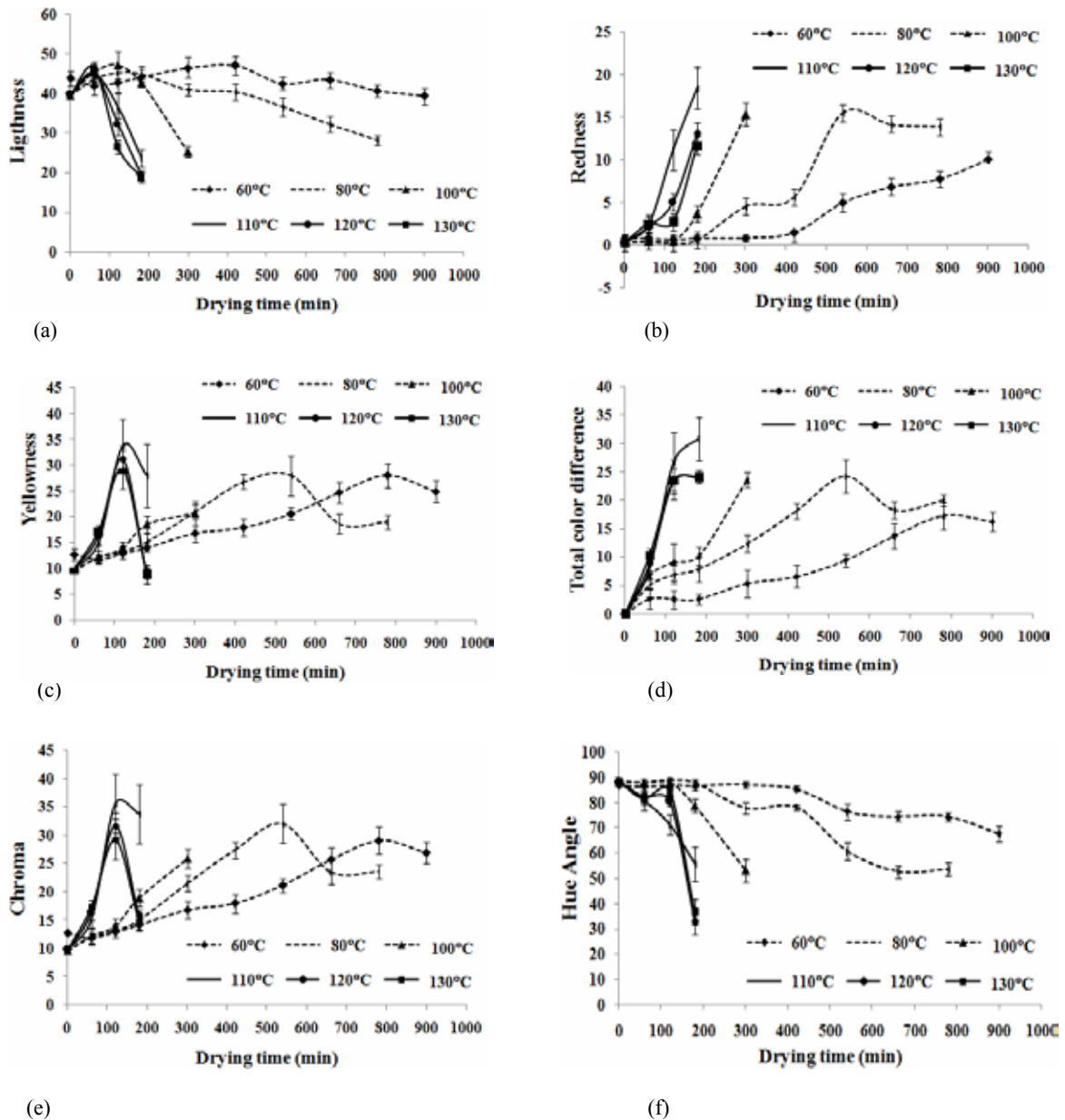


Fig. 2. Color changes in longan flesh drying for various drying temperatures

3.2 Kinetics of color change

Experimental data for changes in parameters L^* , a^* , b^* and ΔE were fitted to different kinetic models Eqs. (5)-(7). The constant parameters of Eq. (8) are shown in Table 1. The value of the coefficients of determination (R^2) was used to select the best fit for the kinetic models. It was observed that the kinetic model 3 (second order polynomial equation) was considered the best model to represent the kinetics of L^*

and b^* . The zero-order kinetic model provided that best fit for the a^* values and this model was an appropriate representation for describing the changes in ΔE values with high R^2 . Fig. 3 and Fig. 4 show the comparison of experimental and predicted values of the color parameters by the proposed models at drying air temperature of 80°C.

Table 1. The results of constant parameters from Eq. (8) at drying temperatures

Color	kinetic parameters	Constant parameters					R^2
		a_1	a_2	a_3	a_4	a_5	
Lightness	k_0	-6.5028×10^{-8}	2.4587×10^{-5}	-3.3719×10^{-3}	0.1994	-4.2901	0.51
	k_1	-	-	8.3547×10^{-7}	-1.1841×10^{-4}	4.2080×10^{-3}	0.45
	A	1.1924×10^{-9}	-4.2430×10^{-7}	5.4593×10^{-5}	-3.0360×10^{-3}	6.1796×10^{-2}	0.84
	B	6.1058×10^{-10}	-2.8924×10^{-7}	4.7579×10^{-5}	-3.1953×10^{-3}	7.5143×10^{-2}	
Redness	k_0	3.0887×10^{-8}	-1.0521×10^{-5}	1.2982×10^{-3}	-6.9660×10^{-2}	1.3691	0.79
	k_1	-	2.5975×10^{-7}	-7.3381×10^{-5}	6.3308×10^{-3}	-0.1746	0.34
	A	-	-1.1462×10^{-6}	3.3478×10^{-4}	-2.9878×10^{-2}	0.8302	0.62
	B	-2.7869×10^{-7}	1.0489×10^{-4}	-1.4461×10^{-2}	0.8634	-18.7817	
Yellowness	k_0	3.8969×10^{-8}	-1.2902×10^{-5}	1.5384×10^{-3}	-7.9144×10^{-2}	1.4783	0.30
	k_1	-	1.2320×10^{-7}	-3.3845×10^{-5}	2.9168×10^{-3}	-8.0532×10^{-2}	0.40
	A	9.7501×10^{-9}	-3.7316×10^{-6}	5.1830×10^{-4}	-3.0992×10^{-2}	0.6732	0.60
	B	-3.5394×10^{-8}	1.3405×10^{-5}	-1.8484×10^{-3}	0.1101	-2.3860	
ΔE	k_0	6.3932×10^{-8}	-2.2530×10^{-5}	2.8665×10^{-3}	0.1579	3.1768	0.84

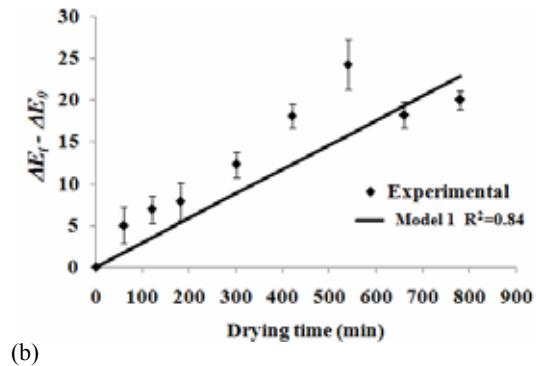
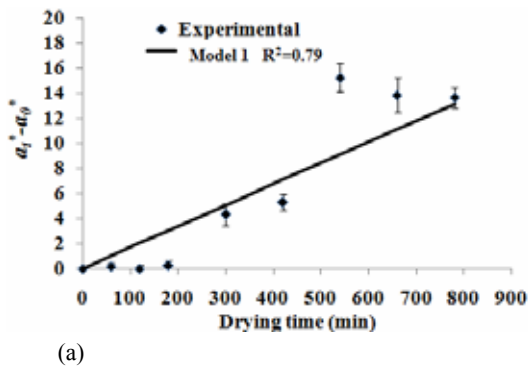


Fig. 3. Comparison of experimental color changes (a^* and ΔE) and predictions by the zero- order kinetic model, Eq.(5), for a drying air temperature of 80°C

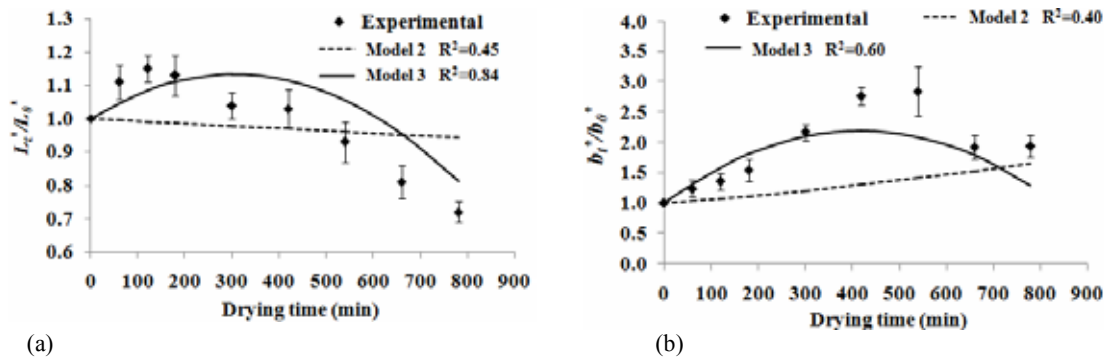


Fig. 4. Comparison of experimental color changes (L^* and b^*) and predictions by the first-order and modified kinetic models, Eq. (6) and Eq. (7), for a drying air temperature of 80°C

4. Conclusion

The color characteristics of longan flesh were studied during air drying using the CIE system determining L^* , a^* and b^* values. The values of L^* and hue angle decreased while the values of a^* , b^* , ΔE and chroma increased and then decreased when the drying time and drying air temperature increased. The modified kinetic model (second order polynomial equation) was the most suitable to describe the changes in L^* and b^* values while the best model describing the changes in a^* values was zero-order kinetic model. The zero-order kinetic model was appropriate representation for describing the changes in ΔE values. Both models could predict the color change of longan flesh during drying relatively well. It was found that the drying air temperature and drying time had an effect on the color change during drying processes.

Acknowledgements

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